Deep Double Descent

Where bigger models, longer training, and more data can hurt

COMP34312 | Mathematical Topics in Machine Learning | Poster Coursework

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From the textbook U-curve to the double-descent peak

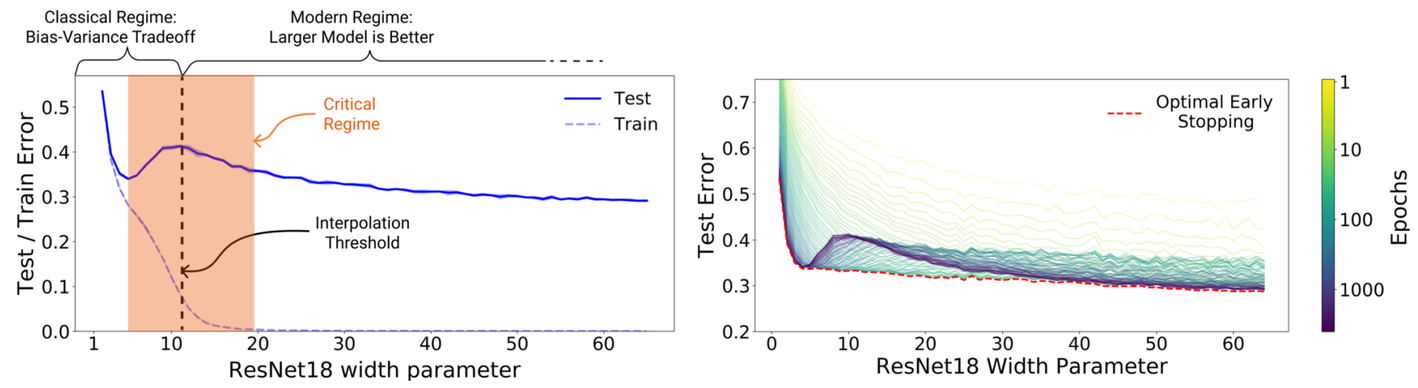


Fig. 1, Nakkiran et al. (2019). Left: test error on ResNet18s, varying width, CIFAR-10 + 15% label noise. **Right:** test error coloured by training epoch; dashed red curve = optimal early stopping.

THE THESIS IN ONE LINE

For modern training procedures the population-risk curve has *two* descents, separated by a sharp peak at the **interpolation threshold**, where the procedure can just fit the train set.

BEYOND SECTION 5.3

The lecture introduces the phenomenon and links it to implicit regularisation. We go further by:

- Replacing *parameter count* with **Effective Model Complexity** (EMC), the capacity of the *whole training procedure* $\mathcal{T}$.
- Showing the second descent appears **model-wise**, **epoch-wise** *and* **sample-wise**.
- Stating the **Generalised Double Descent Hypothesis** and the limits of its current justification.

Setting: the lecture decomposition we extend

Following the course notation (Notes §5.1, Eqs. 68–73):

$$\hat{f} \leftarrow \mathcal{A}(\mathcal{F}; \mathcal{S}), \qquad R(f) := \mathbb{E}_{xy}[\ell(y, f(x))], \qquad \hat{R}(f, \mathcal{S}) := \frac{1}{n}\sum_{i=1}^{n} \ell(y_i, f(x_i)).$$

The excess risk decomposes into three statistically meaningful gaps:

$$\underbrace{R(\hat{f}) - R(y^*)}_{\text{excess risk}} = \underbrace{R(\hat{f}) - R(f_{\text{erm}})}_{\text{optimisation}} + \underbrace{R(f_{\text{erm}}) - R(f^*)}_{\text{estimation}} + \underbrace{R(f^*) - R(y^*)}_{\text{approximation}}.$$

The classical bias–variance trade-off bundles *approximation + part of estimation* into **bias** and the rest into **variance**; what the lecture summary calls *incomplete*, not wrong.

- **Lecture U-curve** (Notes Fig. 17) holds $\mathcal{F}, \mathcal{A}$ fixed and grows $\mathcal{F}$.
- **Paper's pivot:** treat the *whole procedure* $\mathcal{T} = (\mathcal{F}, \mathcal{A}, \text{epochs}, \text{reg.}, \text{labels})$ as the single object whose capacity matters, so optimisation, estimation, and labels jointly determine the peak.

Definition + Hypothesis (Nakkiran et al. 2019)

DEFINITION 1: EFFECTIVE MODEL COMPLEXITY

A **training procedure** $\mathcal{T}$ maps any labelled train set $S = \{(x_i, y_i)\}_{i=1}^n$ to a classifier $\mathcal{T}(S)$. For an error tolerance $\varepsilon > 0$,

$$\boxed{\mathrm{EMC}_{\mathcal{D},\varepsilon}(\mathcal{T}) := \max\left\{ n \;\middle|\; \mathbb{E}_{S\sim\mathcal{D}^n}\left[\mathrm{Error}_S(\mathcal{T}(S))\right] \le \varepsilon \right\}}$$

i.e. the largest sample size on which $\mathcal{T}$ can still drive train error below ε. Note that EMC depends on the optimiser, training time, augmentation and labels, not just on the architecture.

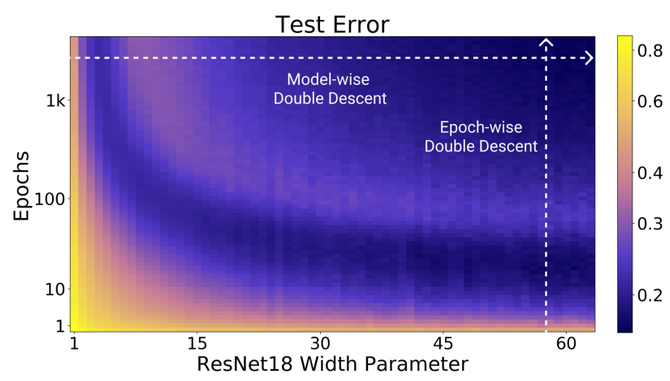
HYPOTHESIS 1: GENERALISED DOUBLE DESCENT

For "natural" $\mathcal{D}$, neural-network procedures $\mathcal{T}$, and small ε (the paper uses $\varepsilon = 0.1$):

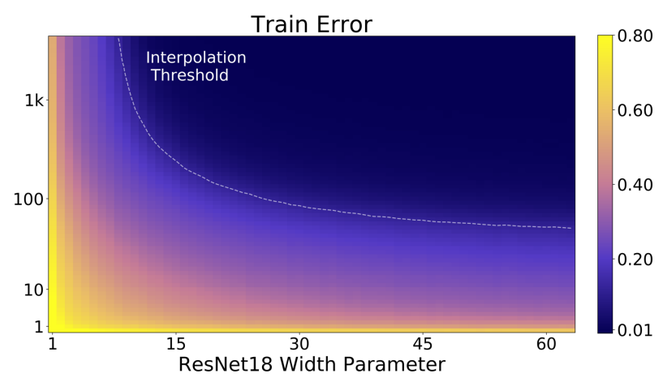
- $\mathrm{EMC} \ll n$ **Under-parameterised:** any change to $\mathcal{T}$ that *increases* EMC *decreases* test error.
- $\mathrm{EMC} \approx n$ **Critically parameterised:** the same change *may decrease or increase* test error. This is the danger zone.
- $\mathrm{EMC} \gg n$ **Over-parameterised:** any change to $\mathcal{T}$ that *increases* EMC *decreases* test error.

Why this is a hypothesis, not a theorem. ε and the words "sufficiently smaller / larger" are heuristic; the critical-interval width is not formally specified for deep nets.

Main empirical evidence: model-wise × epoch-wise double descent



Test error. Bright ridge = peak.



Train error. Dashed = interpolation threshold ($\hat{R} \to 0$).

READING THE HEATMAPS

- **Same axes** for both: ResNet18 width parameter $k \in [1, 64]$ ($k = 64$ is the standard ResNet18) versus training epochs (log).
- **Horizontal slice** ⇒ **model-wise**: peak as width grows.
- **Vertical slice** ⇒ **epoch-wise**: peak as training continues.
- **Key alignment:** the bright *test* ridge (left) sits on top of the *train* interpolation curve (right), so the peak lies *at* the threshold, not far past it.

SETUP (PAPER §4)

ResNet18, CIFAR-10, **15% label noise**, data augmentation, Adam (lr 1e-4), up to 4k epochs. Widths $[k, 2k, 4k, 8k]$ for $k \in [1, 64]$. Identical results hold under SGD and on CIFAR-100; the peak is *also* present without label noise (Fig. 4a, paper).

The unifying variable is EMC, not parameter count. Vary anything in $\mathcal{T}$ that moves EMC across n and the peak moves with it.

Sample-wise: more data can locally hurt

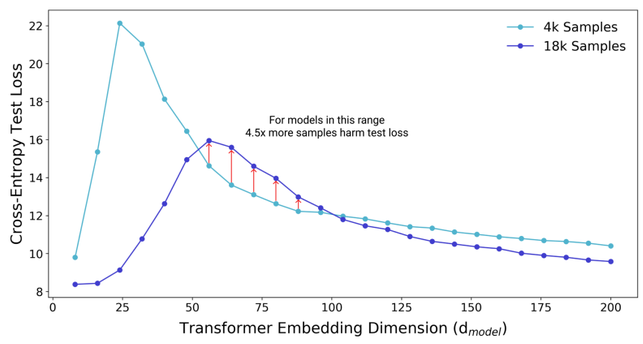


Fig. 3, Nakkiran et al. IWSLT'14 De–En, 6-layer Transformer, $d_{\text{ff}} = 4d_{\text{model}}$, 80k gradient steps, 10% label smoothing. Bars highlight d_{model} range where 4.5× more data *worsens* test loss.

- Fixing $(\mathcal{T}, \mathcal{F})$ and increasing n leaves the interpolation threshold in EMC space unchanged, but *shifts* n relative to it.
- Widths near the small-sample threshold get *pushed into* the critical regime by adding data.
- Local non-monotonicity only: asymptotically, more data still wins.

Mechanism (intuition; rigorous only for linear models)

CRITICALLY PARAMETERISED

Effectively *one* interpolating model. Forcing it to fit any noisy/misspecified label *destroys global structure* ⇒ test error spikes.

OVER-PARAMETERISED

Many interpolants. The optimiser's **implicit bias** (e.g. SGD toward small-norm) lets it "absorb" noise while keeping the on-distribution map smooth.

WHERE THIS IS RIGOROUS

Mei & Montanari (2019), Belkin et al. (2019) prove the test-risk peak for overparameterised linear regression with mis-specification; Bartlett et al. (2020) prove benign overfitting under min-norm interpolation. The deep-net analogue is conjectured, not proved.

What we do *not* claim

- "Bias–variance is wrong." **No**: it is correct within a fixed class, but *incomplete* once $\mathcal{T}$ varies.
- "Bigger/longer/more data is always better." **No**: each can be *worse* in the critical interval.
- "Label noise causes the peak." Noise *amplifies* it; peak appears at 0% noise too (Figs. 4a, 7) under model mis-specification.
- "Parameters = capacity." Parameters are one input to EMC; epochs and regularisation matter as much.
- "EMC is a finished theory." ε and "sufficiently smaller/larger" are heuristic; deep-net mechanism is conjectured.

Practical message & open questions

FOR PRACTITIONERS

- **Validate**: train and test curves *uncouple* near the interpolation threshold.
- **Avoid the critical regime**: early stopping, weight decay, ensembling (Fig. 28), or scale past the threshold.
- **Data augmentation shifts the peak right** (Fig. 5): a free way to escape it.

OPEN SCIENTIFIC QUESTIONS

- Principled ε; closed-form critical-interval width.
- A non-linear EMC theorem for over-parameterised nets.
- SGD implicit bias for realistic architectures and losses.

Interpolation is a threshold to understand, not a final verdict.

KEY REFERENCES

[1] Nakkiran, Kaplun, Bansal, Yang, Barak, Sutskever. *Deep Double Descent: Where Bigger Models and More Data Hurt.* arXiv:1912.02292 (2019). [2] Belkin, Hsu, Ma, Mandal. *Reconciling modern ML practice and the classical bias–variance trade-off.* PNAS 116(32) (2019). [3] Zhang, Bengio, Hardt, Recht, Vinyals. *Understanding deep learning requires rethinking generalization.* ICLR (2017). [4] Mei & Montanari. *The generalization error of random features regression.* Comm. Pure Appl. Math. (2022). [5] Bartlett, Long, Lugosi, Tsigler. *Benign overfitting in linear regression.* PNAS 117(48) (2020). [6] Brown & Mukherjee. *COMP34312 Notes 2026, §5.*

AI-USE DECLARATION

Anthropic Claude was used as a drafting assistant for layout and prose tightening, and to double-check notation against the cited paper and the COMP34312 notes. All claims and figures are verified by the authors against primary references. Figures 1–3 reproduced from Nakkiran et al. (2019).